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Course code:CSA0613

Course title: Design and Analysis of Algorithms

CO4_AT1- Code Challenge

Title: Maximum Sum Subarray Using Dynamic Programming

Aim

To find the contiguous subarray with the maximum possible sum using Dynamic Programming.

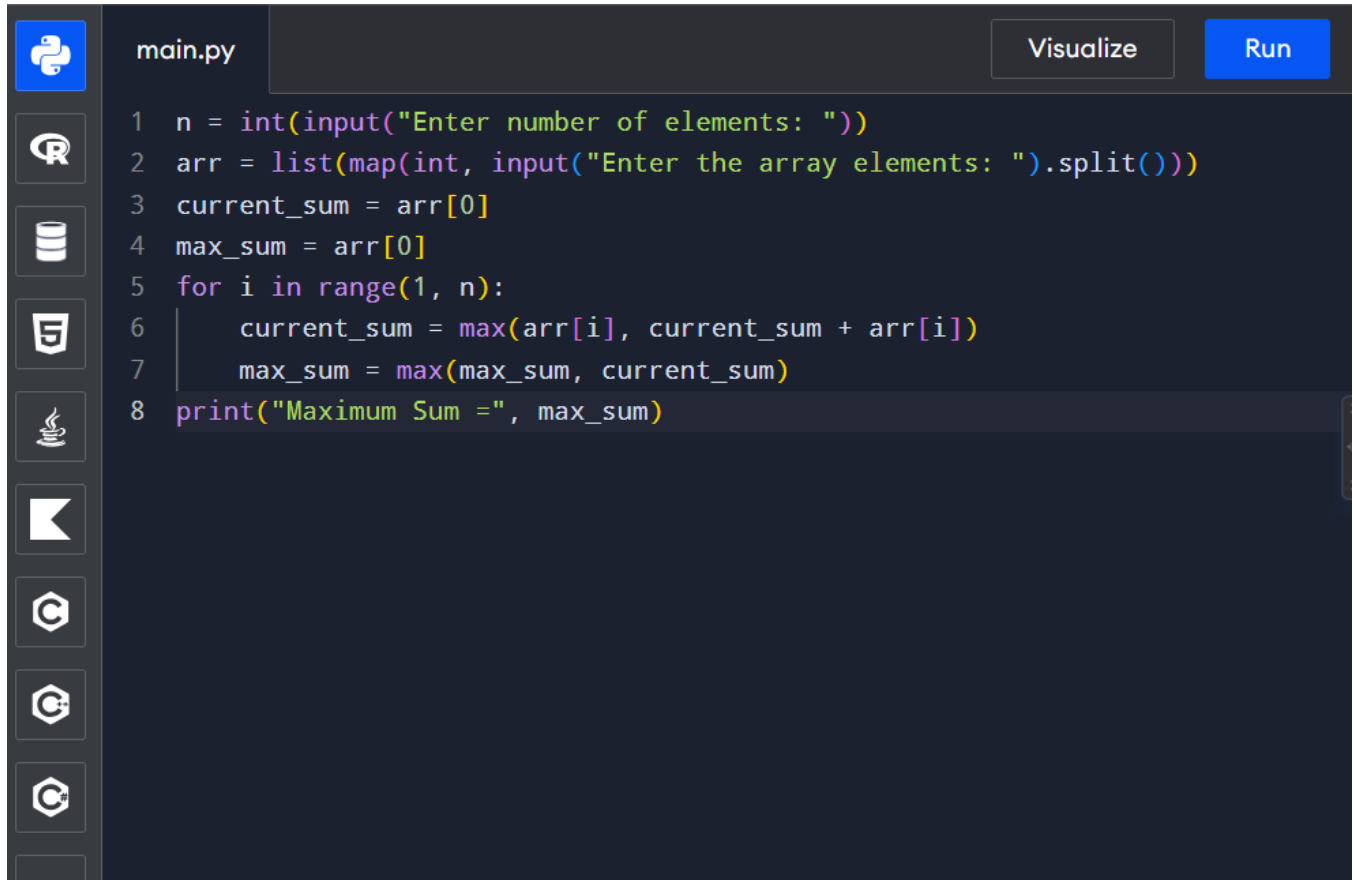
Software Used

- Python

Algorithm

1. Start the program.
2. Read the number of elements.
3. Read the array elements.
4. Set the first element as the current sum and maximum sum.
5. Check each remaining element one by one.
6. Decide whether to start a new subarray or add the element to the existing subarray.
7. Update the current sum.
8. Update the maximum sum if a larger sum is found.
9. Display the maximum sum.
10. Stop the program.

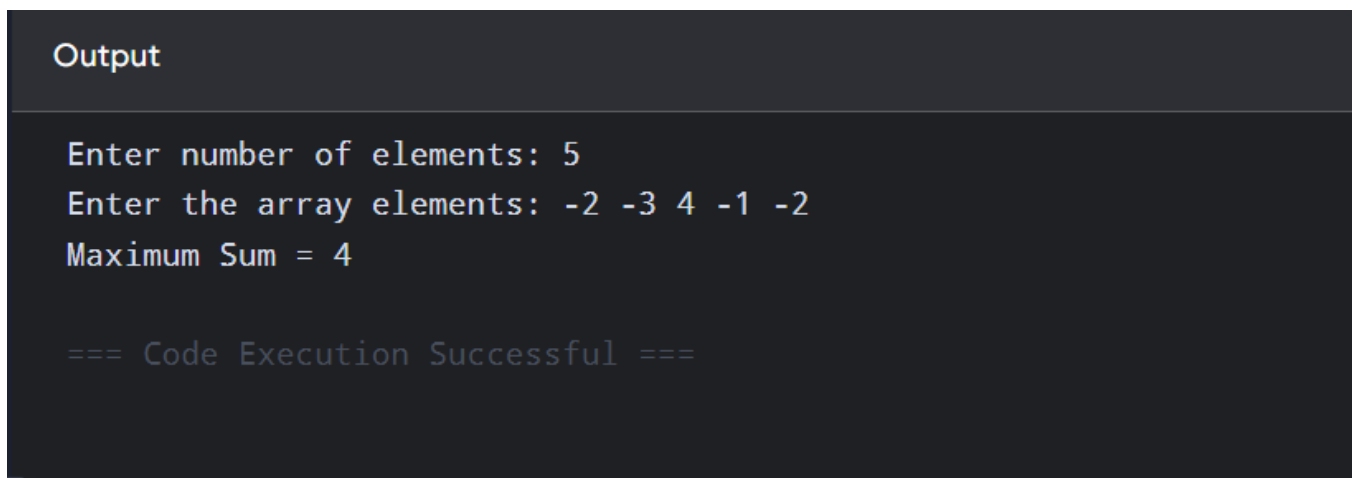
Source Code (Python)



The image shows a Python IDE interface. On the left is a vertical toolbar with icons for Python, R, a database, a trash can, a coffee cup, a left arrow, and several circular icons. The main area is a code editor with a dark background. At the top left of the editor is a tab labeled 'main.py'. At the top right are two buttons: 'Visualize' and 'Run'. The code in the editor is as follows:

```
1 n = int(input("Enter number of elements: "))
2 arr = list(map(int, input("Enter the array elements: ").split()))
3 current_sum = arr[0]
4 max_sum = arr[0]
5 for i in range(1, n):
6     current_sum = max(arr[i], current_sum + arr[i])
7     max_sum = max(max_sum, current_sum)
8 print("Maximum Sum =", max_sum)
```

OUTPUT



The image shows an 'Output' window with a dark background. It displays the text entered by the user and the program's output. At the bottom, it shows a success message.

```
Output

Enter number of elements: 5
Enter the array elements: -2 -3 4 -1 -2
Maximum Sum = 4

=== Code Execution Successful ===
```

Result

Thus, the program to find the Maximum Sum Subarray using Dynamic Programming was implemented successfully.

